Ex. No. 4 — Strategize Impedance Matching with Characteristic Impedance and Load Impedance through the Measurement of a) Location of the Stub b) Length of the Stub

Test Case 1 — Varying Load Impedance (Z_R)

Optimize impedance matching using single-stub matching by finding the location and length of the stub for load impedance values from 100∠30° to 350∠30° Ω, in steps of 50 Ω, keeping the characteristic impedance Z_o = 60 Ω constant.

Scilab Program

```
clear; clear; clc;
ZR = 100; Zo = 60; f = 100*10^6;    // set ZR (or Zo) per row
lo = 300/(f*10^-6);                 // lo = wavelength
Ls = (lo/(2*%pi))*(atan(sqrt(ZR/Zo)));
printf('-Point of attachment = %f cms\n', round(Ls*10^2*10)/10)
Lt = (lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
printf('-Length of the short circuited stub = %f cms', round(Lt*10^2));
```

Result Table (Z_o = 60 Ω, f = 100 MHz, variable Z_R∠30°)

S.No	Z■ (ohms) ∠30°	Point of Attachment (cm)	Stub Length (cm)
1	100	43.5	202
2	150	48.1	189
3	200	51.1	182
4	250	53.2	177
5	300	54.9	174
6	350	56.3	172

Test Case 2 — Varying Characteristic Impedance (Z_o)

Optimize impedance matching using single-stub matching by finding the location and length of the stub for characteristic impedance values from 600 Ω to 1800 Ω, in steps of 200 Ω, keeping the load impedance $Z_R = 100\angle 30°$ Ω constant.

Scilab Program

```
clear; clear; clc;
ZR = 100; Zo = 60; f = 100*10^6;    // set ZR (or Zo) per row
lo = 300/(f*10^-6);                 // lo = wavelength
Ls = (lo/(2*%pi))*(atan(sqrt(ZR/Zo)));
printf('-Point of attachment = %f cms\n', round(Ls*10^2*10)/10)
Lt = (lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
printf('-Length of the short circuited stub = %f cms', round(Lt*10^2));
```

Result Table   ($Z_R = 100\angle 30°$ Ω, f = 100 MHz, variable Z_o)

S.No	Z■ (ohms)	Point of Attachment (cm)	Stub Length (cm)
1	600	18.5	128
2	800	16.2	132
3	1000	14.6	134
4	1200	13.4	135
5	1400	12.5	137
6	1600	11.7	138
7	1800	11.1	138

Result

Single-stub impedance matching was analyzed for a range of load impedance values (with Z_o fixed at 60 Ω) and for a range of characteristic impedance values (with Z_R fixed at $100\angle 30°$ Ω), and in each case the point of attachment and the length of the short-circuited stub required for matching were determined.